# Probability and Statistic (for CS Students)

## Lecture 12: Continous Random Variables I

University of Mazandaran

Fall 1405

# Recap

$X \sim \text{Bern}(p)$
Successes in one trial

$X \sim \text{Geo}(p)$
Trials until one success

$X \sim \text{Poi}(\lambda)$
Events in one time interval

$X \sim \text{Bin}(n, p)$
Successes in $n$ trials

$X \sim \text{NegBin}(r, p)$
Trials until $r$ success

# Continuous Random Variables

Not all values are discrete

How can we model continuous values?

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How about the probability distribution of the amount of time until the next earthquake.

# Riding a Bus

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You show up at $2:15$ pm.

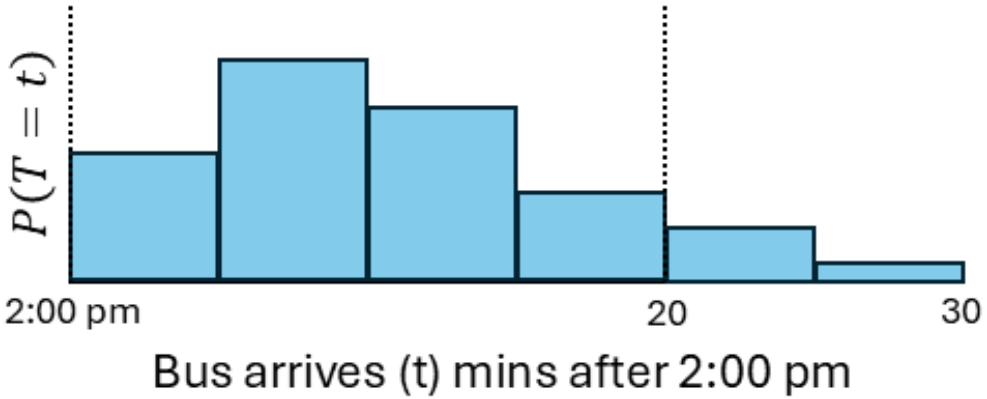
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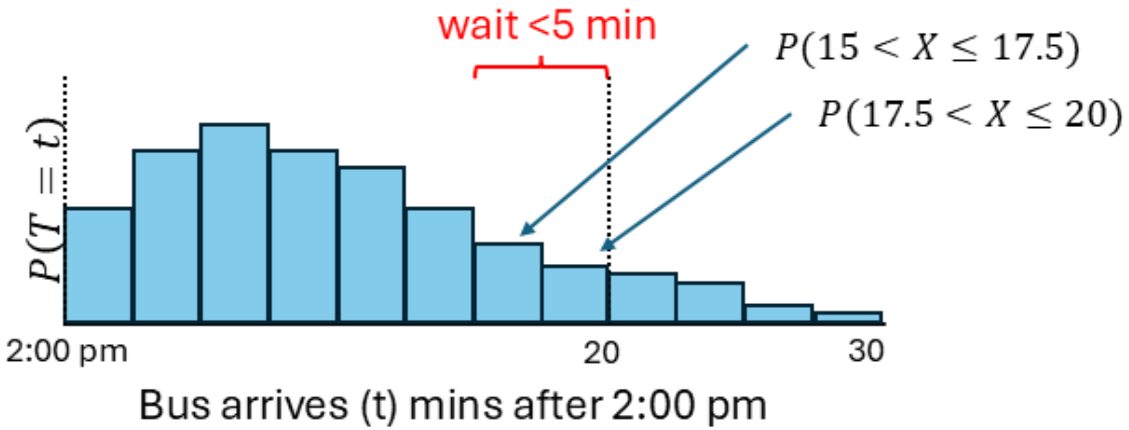


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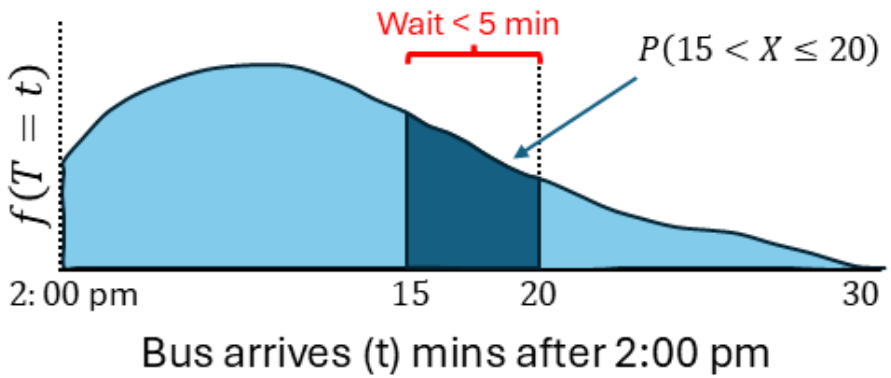


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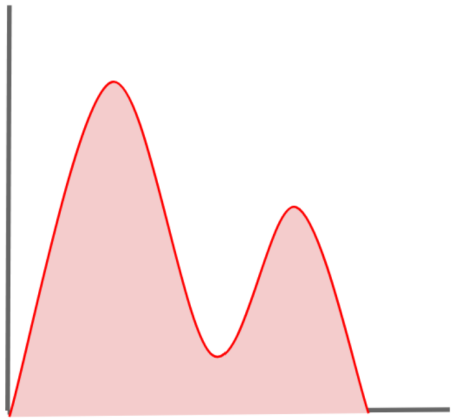
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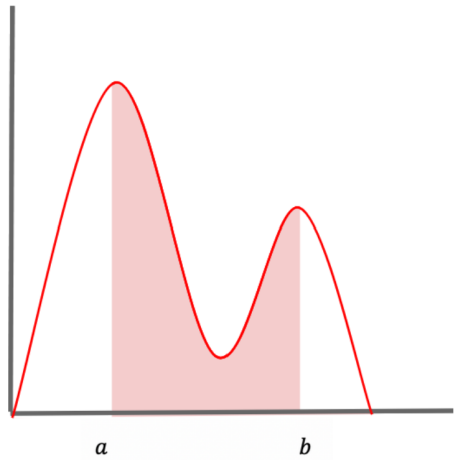
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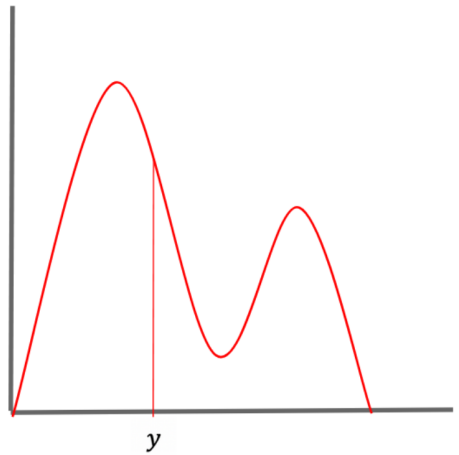
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  - the likelihood is non-zero!

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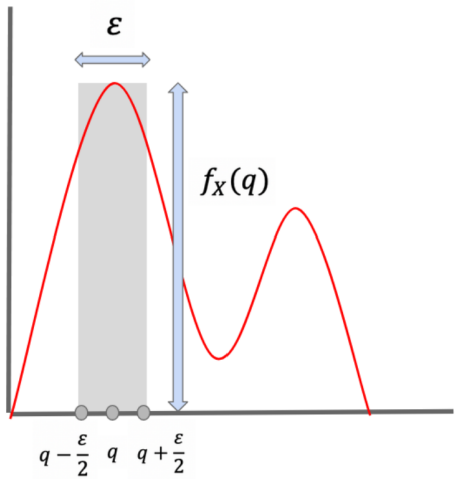
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# Properties of PDFs

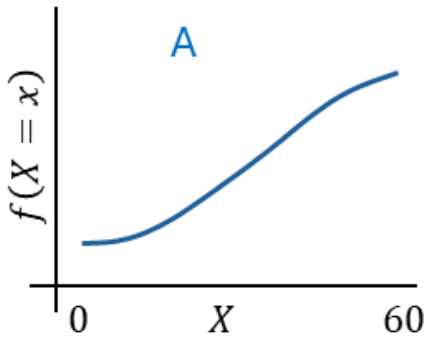
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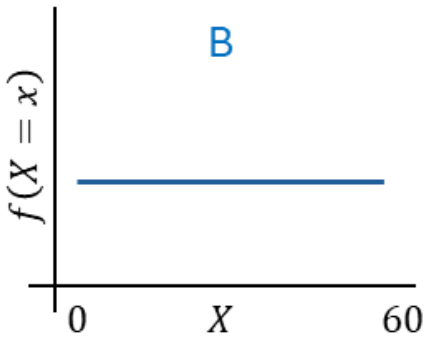
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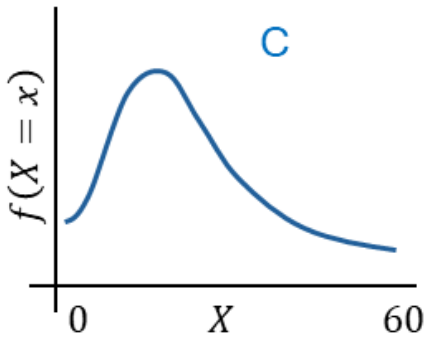
A probability

# Properties of PDFs

probability density functions articulate relative belief.
Let $X$ be a random variable which is the # of minutes after 2pm that the bus arrives at the stop:

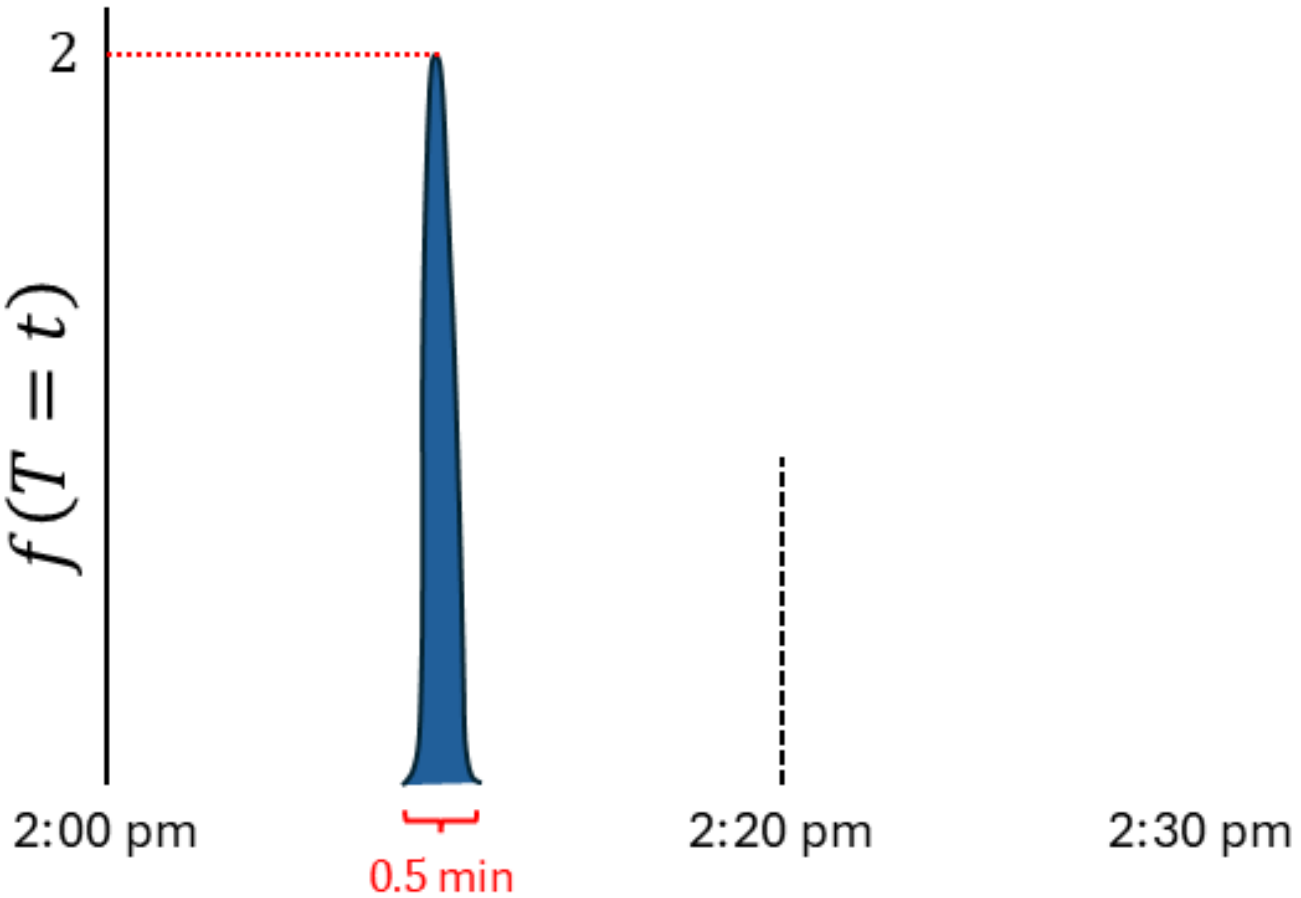



which of these represent that you think the arrival is more likely to be close to 3:00 pm?

# Probability vs Likelihood



# Probability vs Likelihood

$f(X = x)$ is not a probability!

# $f(X = x)$ vs $P(X = x)$

The probability that a discrete random variable $X$ takes on the value $x$:

$$P(X = x)$$

The derivative of the probability that a continuous random variable $X$ takes at the value $x$:

$$f(X = x)$$

# $f(X = x)$ vs $P(X = x)$

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Probability density function is sometimes called likelihood function.

# The `random()` Function

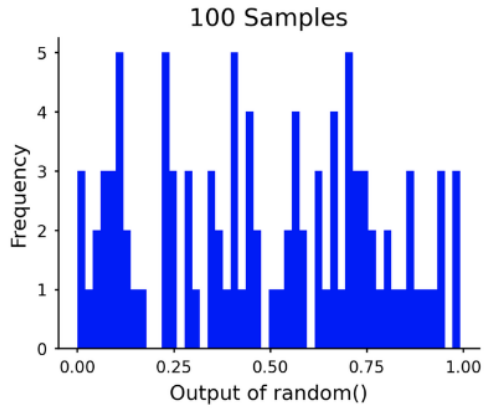
- Outputs values in $[0, 1]$
- All possible values are equally likely
- This is a continuous rv

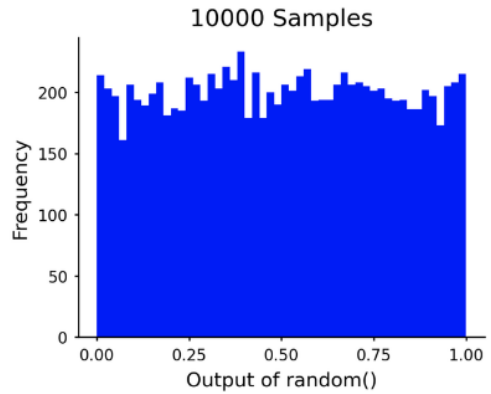
```python
import random

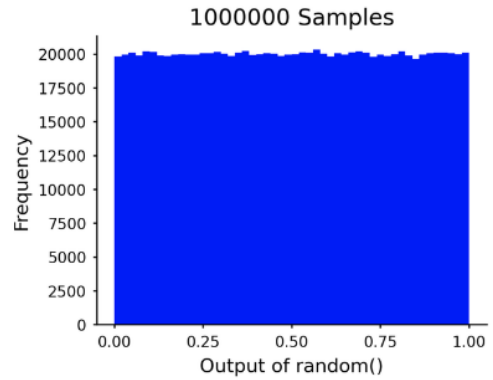
samples_small = []
for i in range(100):
    samples_small.append(random.random())

samples_medium = []
for i in range(10000):
    samples_medium.append(random.random())

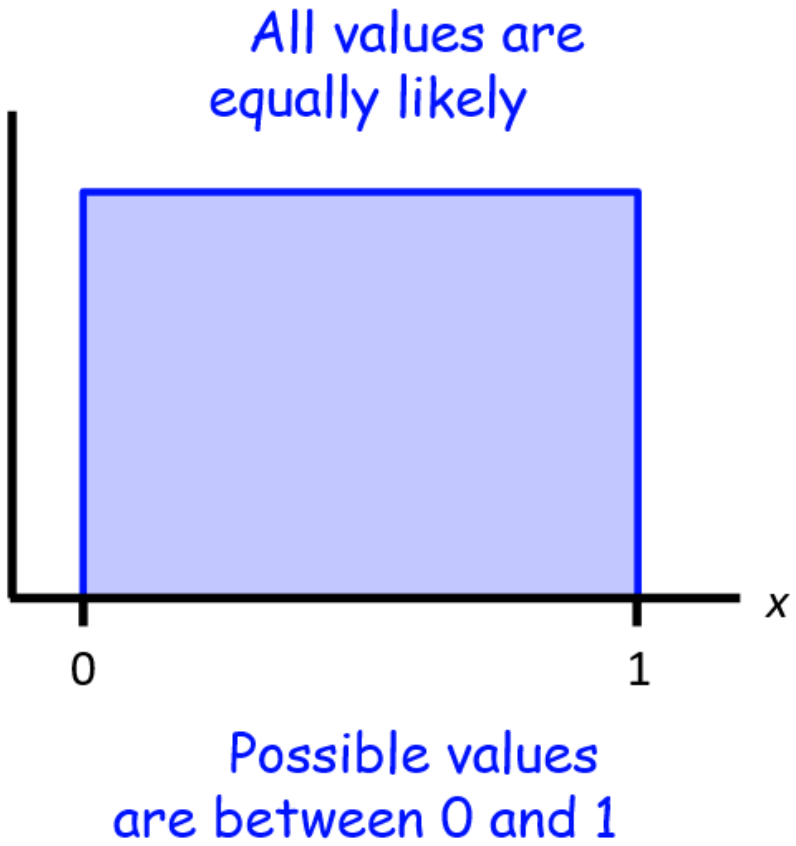
samples_large = []
for i in range(1000000):
    samples_large.append(random.random())
```



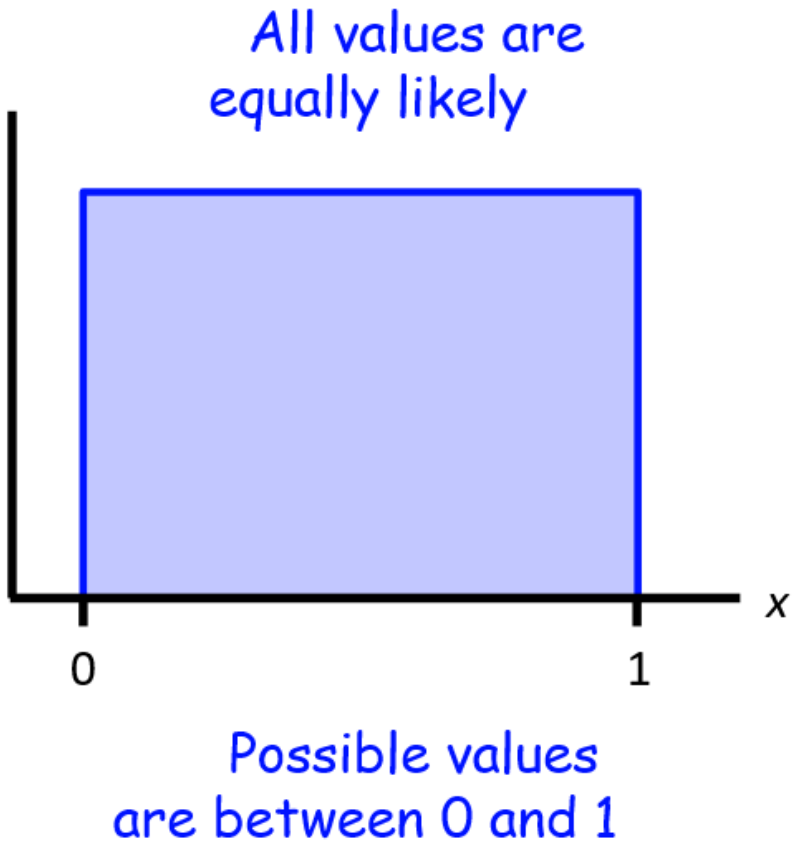



# $X \sim \text{Uniform}(0, 1)$: A Continuous Random Variable

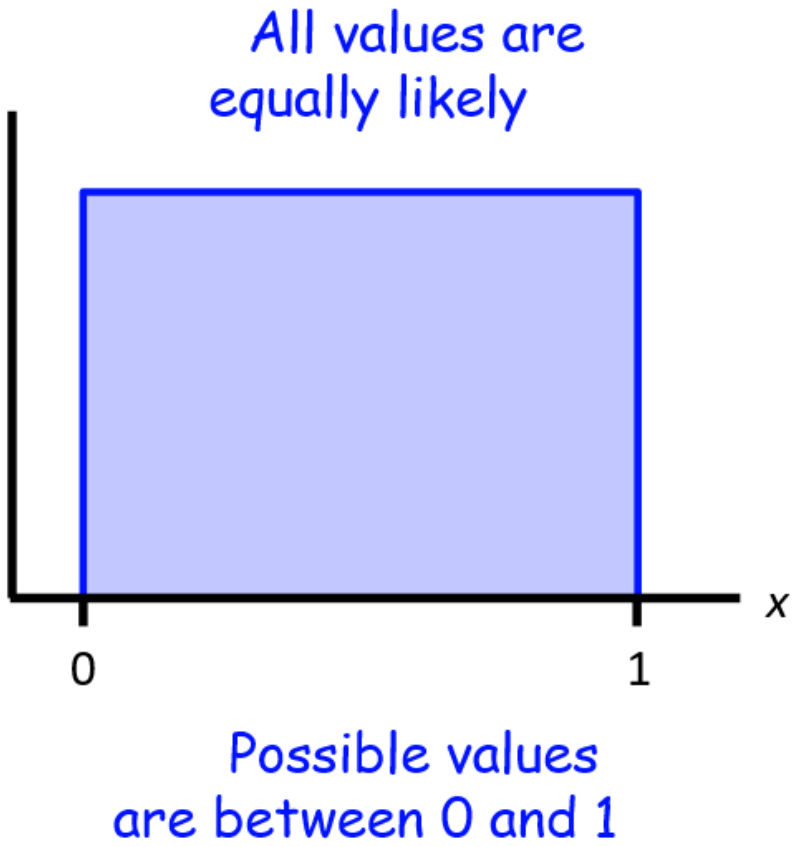


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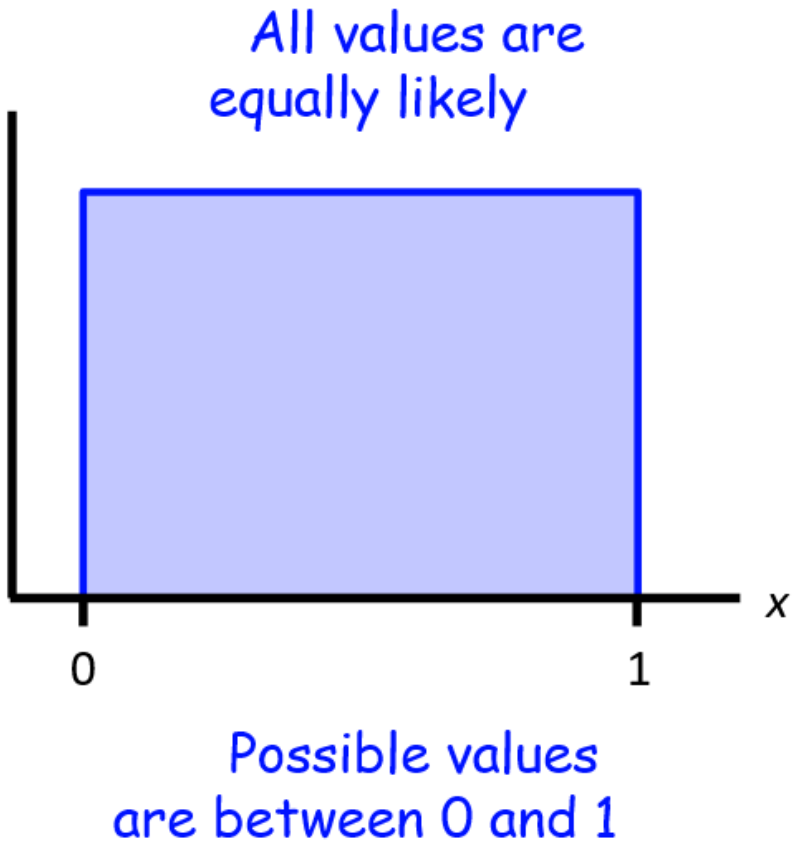

$$P(0 \leq X \leq 1) = ?$$

# $X \sim \text{Uniform}(0, 1)$: A Continuous Random Variable




$$P(0 \leq X \leq 1) = 1$$

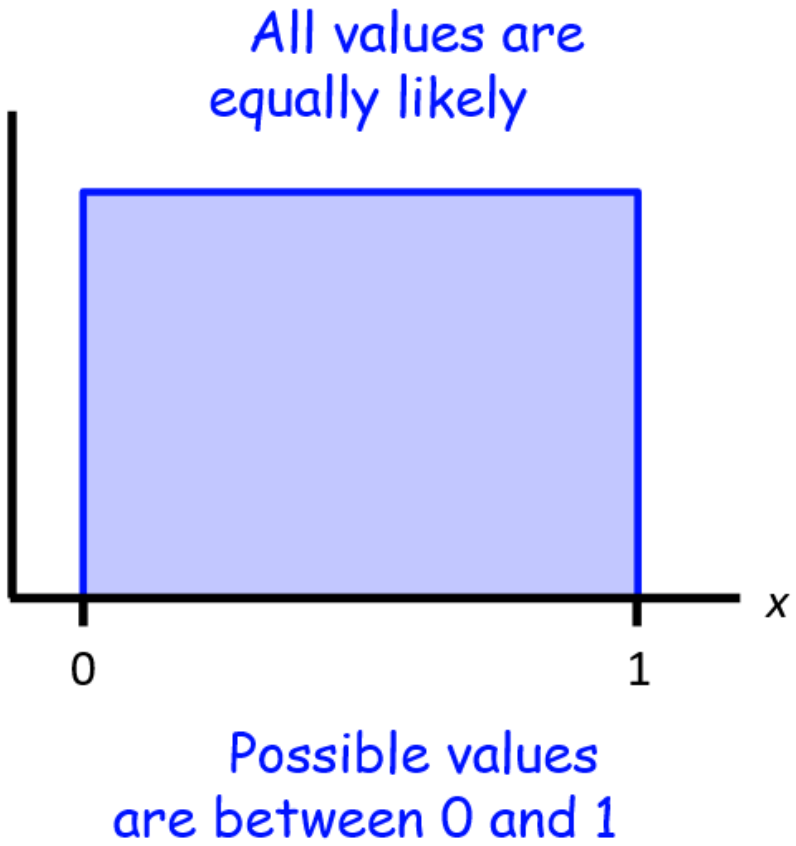
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$$P(0 \leq X \leq 1) = 1$$

Probability of the whole sample space must equal 1 (Axiom 2)

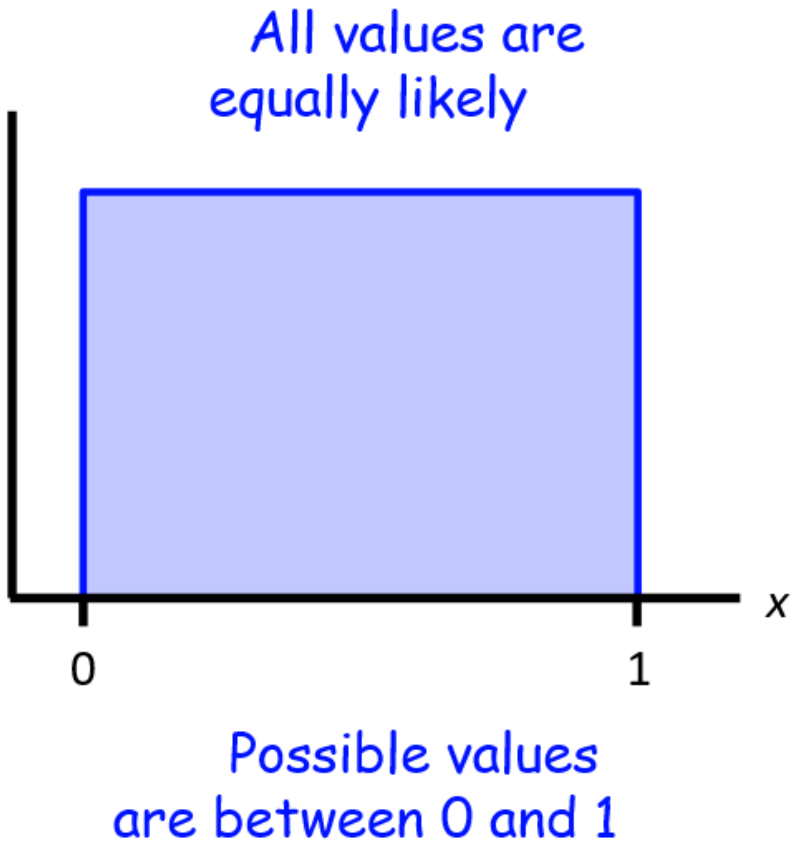
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$$P(0 \leq X \leq 1) = 1$$

$$P(0.5 \leq X \leq 1) = ?$$

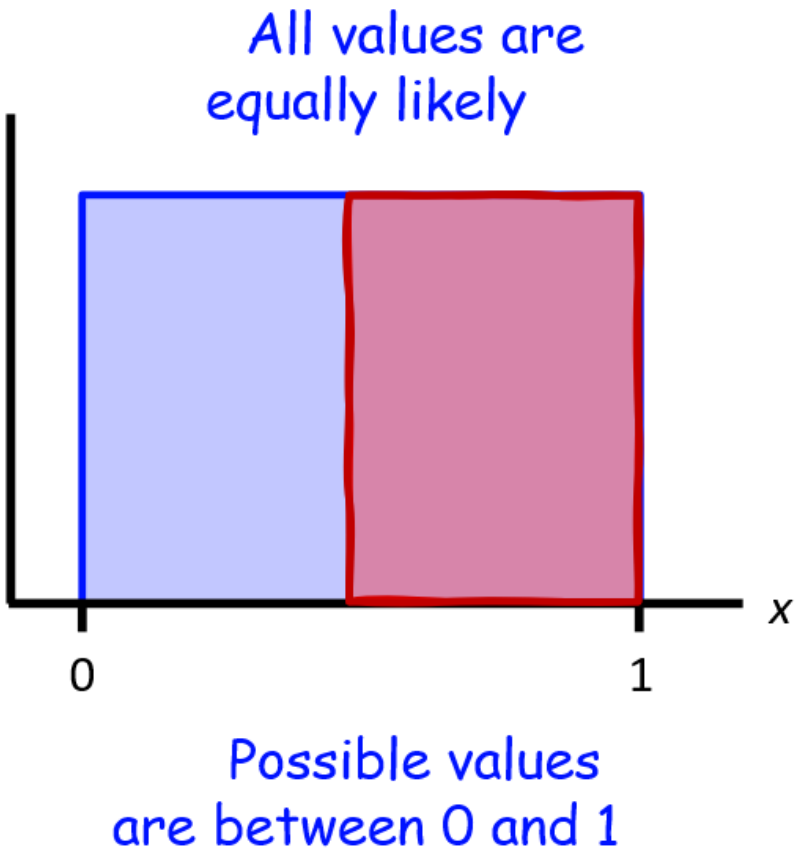
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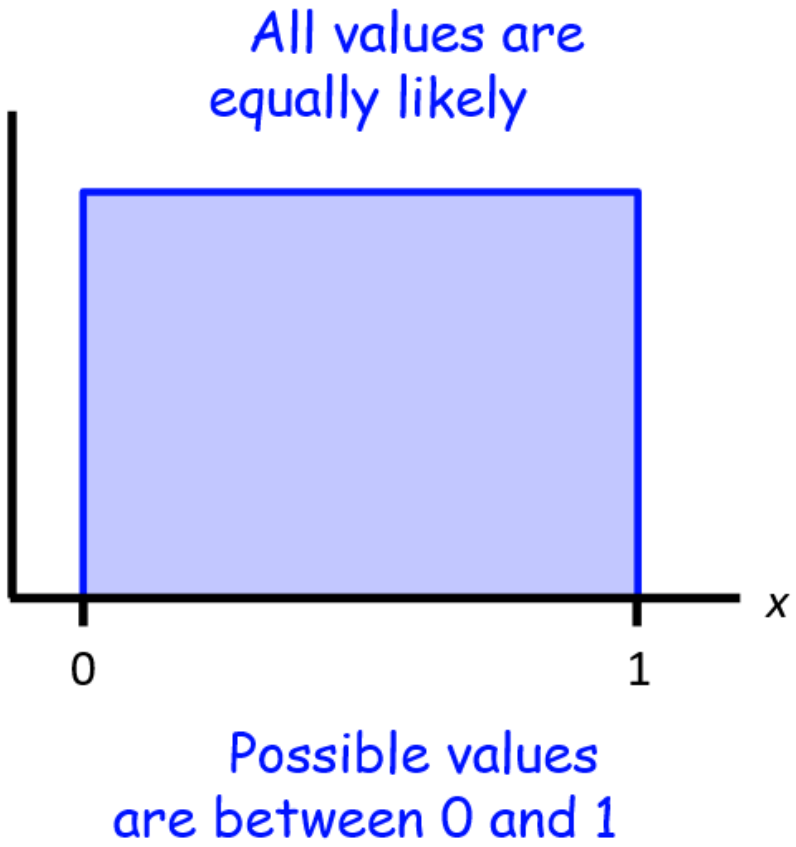



$$P(0 \leq X \leq 1) = 1$$

$$P(0.5 \leq X \leq 1) = 0.5$$

Half of all possible outcomes are between 0.5 and 1

# $X \sim \text{Uniform}(0, 1)$: A Continuous Random Variable

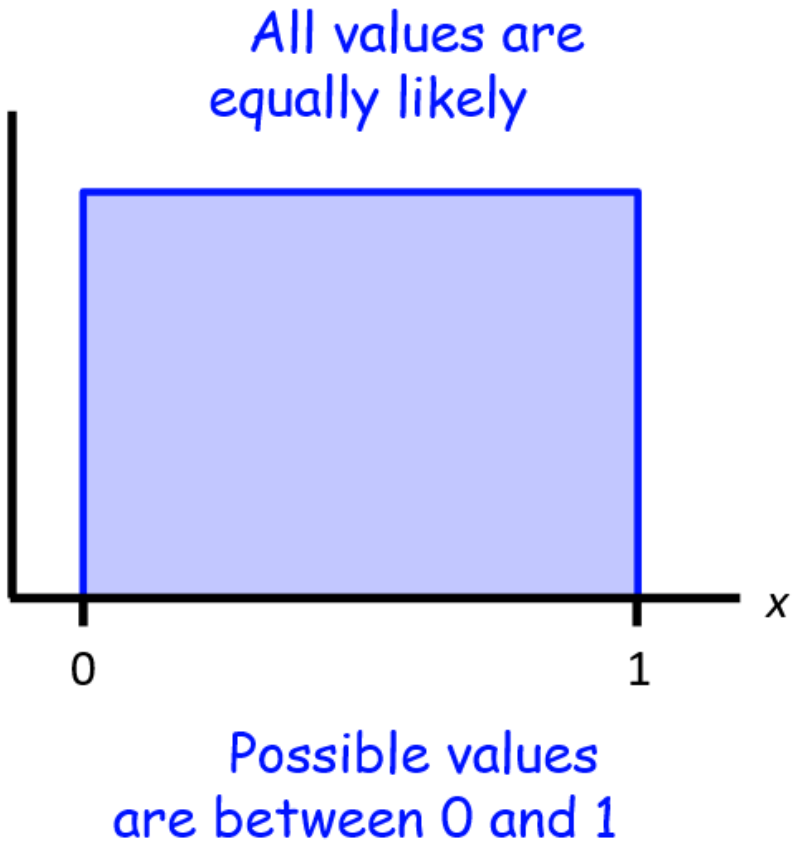



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$$P(0.5 \leq X \leq 0.6) = ?$$

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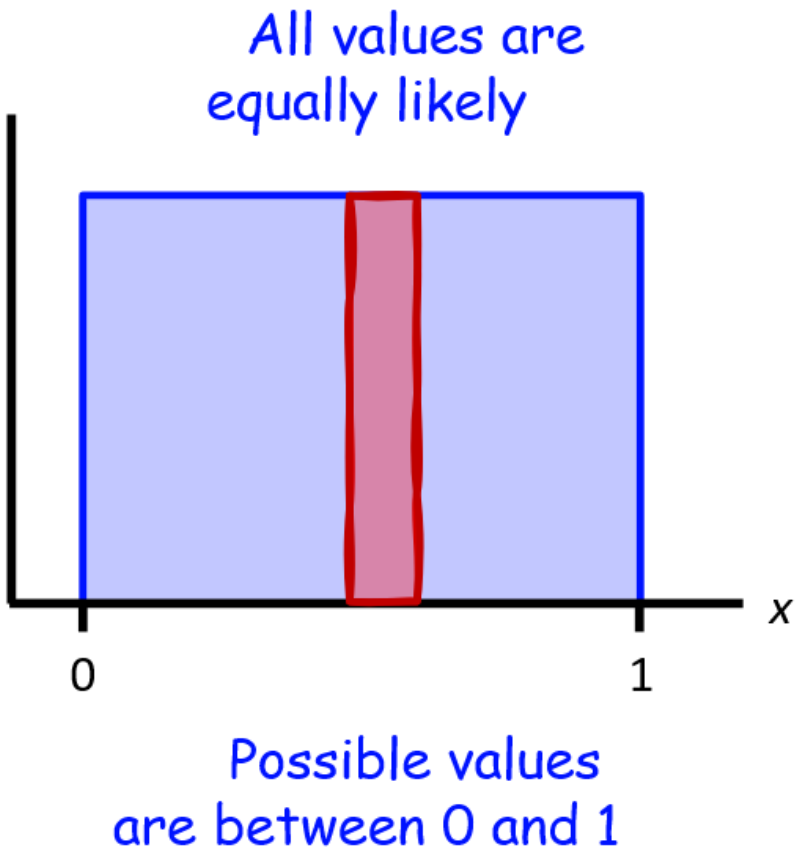



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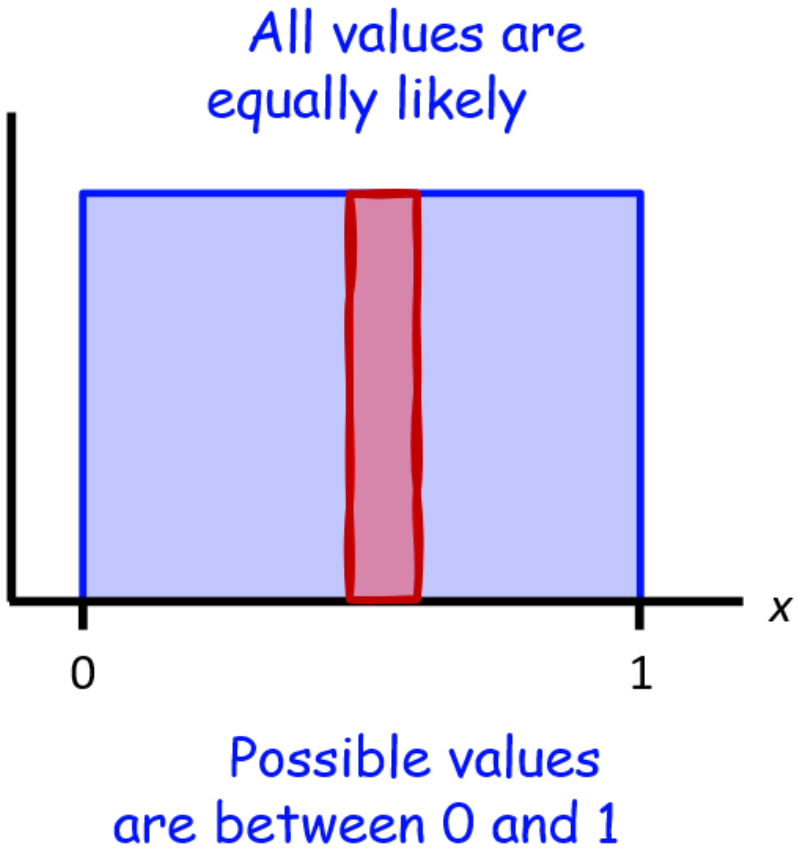
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$$P(0 \leq X \leq 1) = 1$$
$$P(0.5 \leq X \leq 1) = 0.5$$
$$P(0.5 \leq X \leq 0.6) = 0.1$$

1/10 of all possible outcomes are between 0.5 and 0.6

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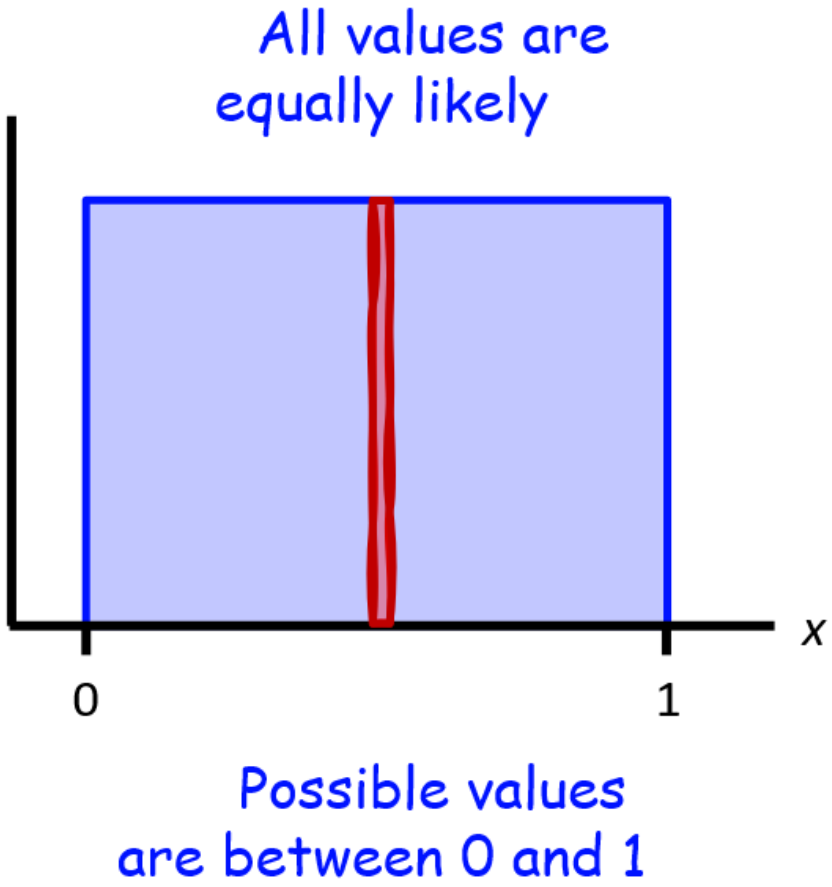



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So far, the pattern looks like:

$$P(\text{start} \leq X \leq \text{end}) = \text{end} - \text{start}$$

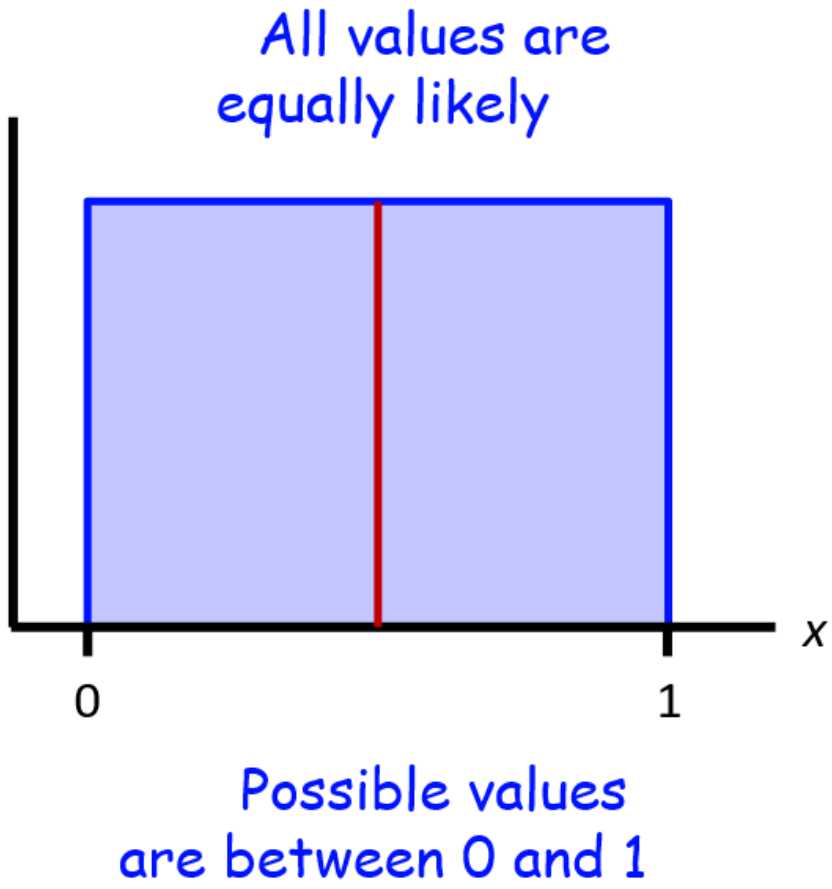
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$$P(0.5 \leq X \leq 0.6) = 0.1$$
$$P(0.5 \leq X \leq 0.5001) = 0.001$$

As we get more precise,
probabilities keep shrinking ...

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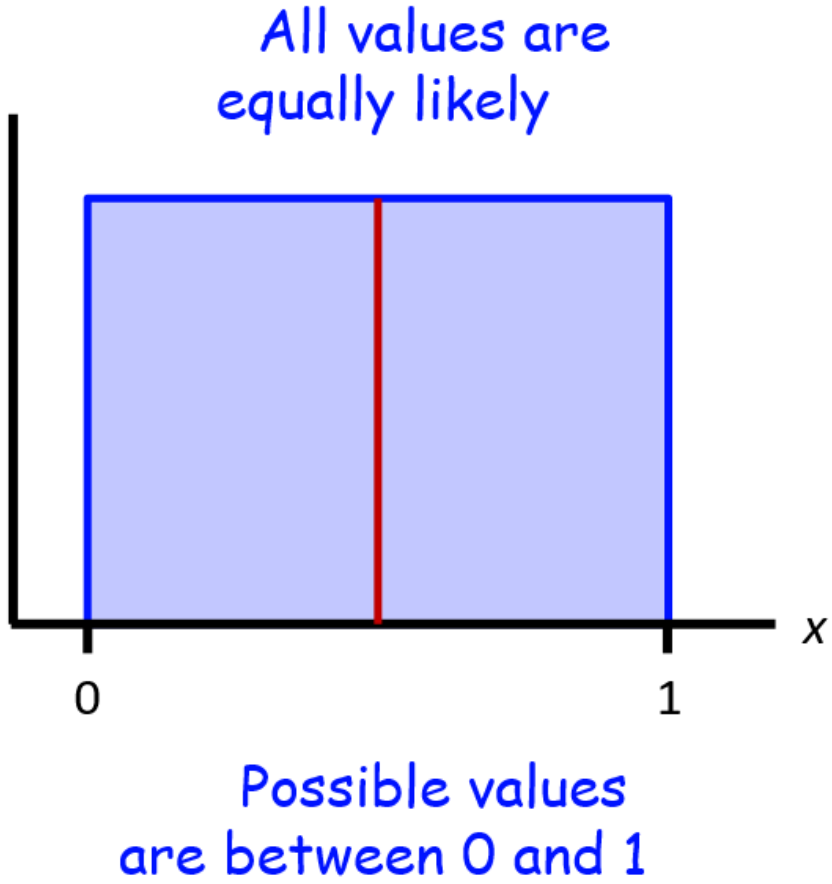
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$$P(0.5 \leq X \leq 0.5001) = 0.001$$

$$P(X = 0.5) = ?$$

# $X \sim \text{Uniform}(0, 1)$: A Continuous Random Variable




$$P(0 \leq X \leq 1) = 1$$
$$P(0.5 \leq X \leq 1) = 0.5$$
$$P(0.5 \leq X \leq 0.6) = 0.1$$
$$P(0.5 \leq X \leq 0.5001) = 0.001$$
$$P(X = 0.5) = 0$$

The probability of any exact outcome,
with infinite precision is zero!

# More General Case

A uniform random variable is equally likely to be any value in an interval.

$$X \sim \text{Uni}(\alpha, \beta)$$

Probability Density

$$f(X = x) = \begin{cases} \frac{1}{\beta - \alpha} & \alpha \leq x \leq \beta \\ 0 & \text{otherwise} \end{cases}$$

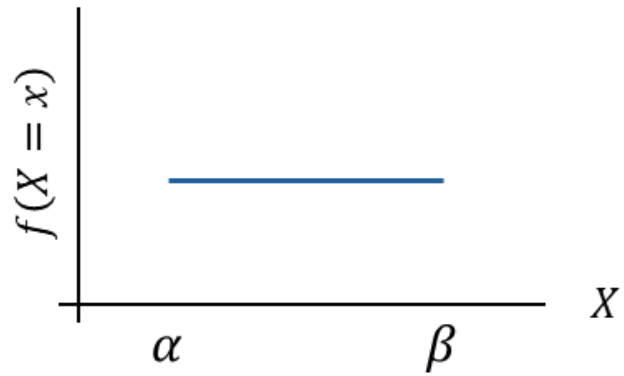
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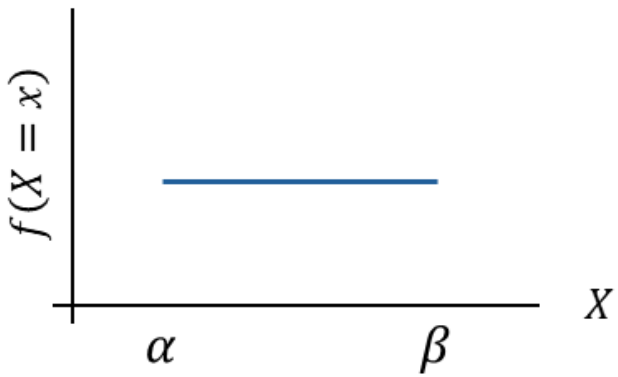
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$$f(X = x) = \begin{cases} \frac{1}{\beta - \alpha} & \alpha \leq x \leq \beta \\ 0 & \text{otherwise} \end{cases}$$




Properties:

$$E[X] = \frac{\beta - \alpha}{2}$$

$$\text{Var}(X) = \frac{(\beta - \alpha)^2}{12}$$

# More General Case

**Notation:** $X \sim \text{Uni}(\alpha, \beta)$

**Description:** A continuous random variable that takes on values, with equal likelihood, between $\alpha$ and $\beta$

**Parameters:** $\alpha \in \mathbb{R}$, the minimum value of the variable.
$\beta \in \mathbb{R}$, $\beta > \alpha$, the maximum value of the variable.

**Support:** $x \in [\alpha, \beta]$

**PDF equation:**
$$f(x) = \begin{cases} \frac{1}{\beta - \alpha} & \text{for } x \in [\alpha, \beta] \\ 0 & \text{else} \end{cases}$$

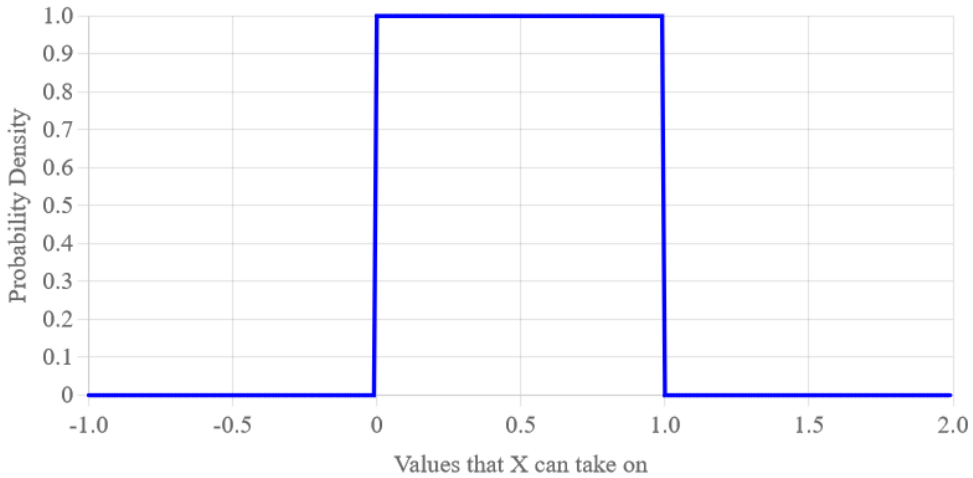
**CDF equation:**
$$F(x) = \begin{cases} \frac{x - \alpha}{\beta - \alpha} & \text{for } x \in [\alpha, \beta] \\ 0 & \text{for } x < \alpha \\ 1 & \text{for } x > \beta \end{cases}$$

**Expectation:** $\text{E}[X] = \frac{1}{2}(\alpha + \beta)$

**Variance:** $\text{Var}(X) = \frac{1}{12}(\beta - \alpha)^2$

**PDF graph:**

Parameter $\alpha$: 0 Parameter $\beta$: 1

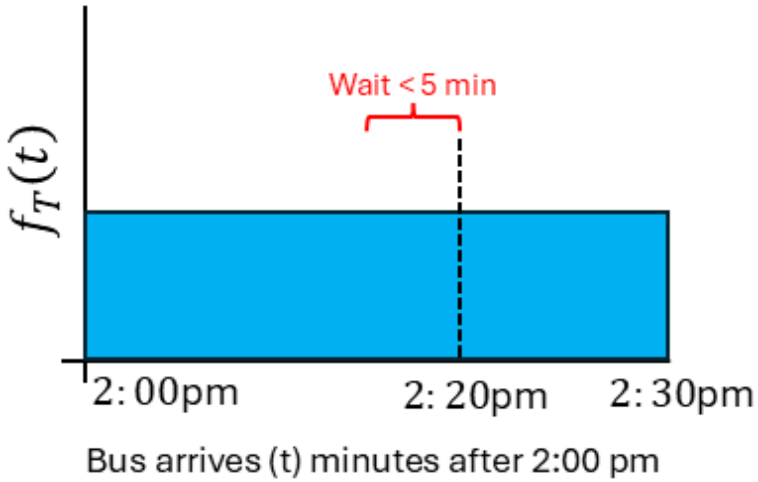


# More General Case

You are running to the bus stop. You don't know exactly when the bus arrives. You believe all times between 2 and 2:30 are equally likely.

You show up at 2:15 pm. What is $P\,(\text{wait} < 5 \text{ minutes})$?

$$T \sim \text{Uni}\,(\alpha = 0, \beta = 30)$$

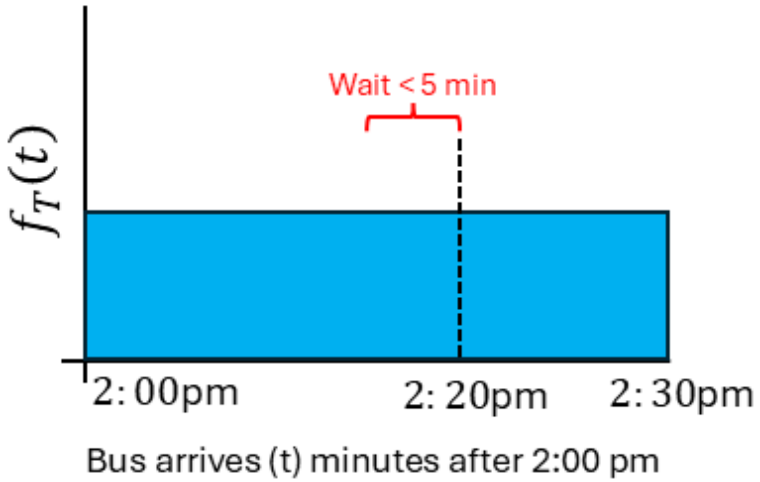


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$$P(\text{wait} < 5) = \int_{15}^{20} \frac{1}{\beta - \alpha} dx$$

$$= \frac{x}{\beta - \alpha}\Big|_{15}^{20}$$

$$= \frac{x}{30 - 0}\Big|_{15}^{20} = \frac{5}{30}$$